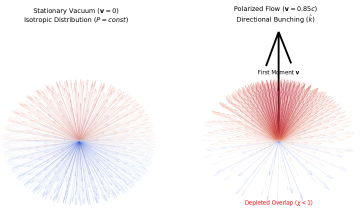
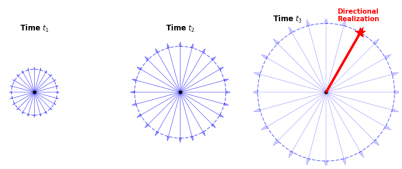
The Null-Directional Substrate: A Visual Guide

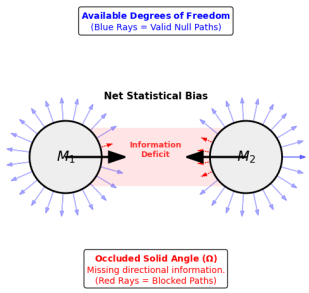
1. The Null-Directional Framework (Structure, Propagation, Gravity)



A. Structure: The vacuum is modeled as a null-directional ensemble (left). Motion corresponds to directional polarization (aberration) of the distribution (right).

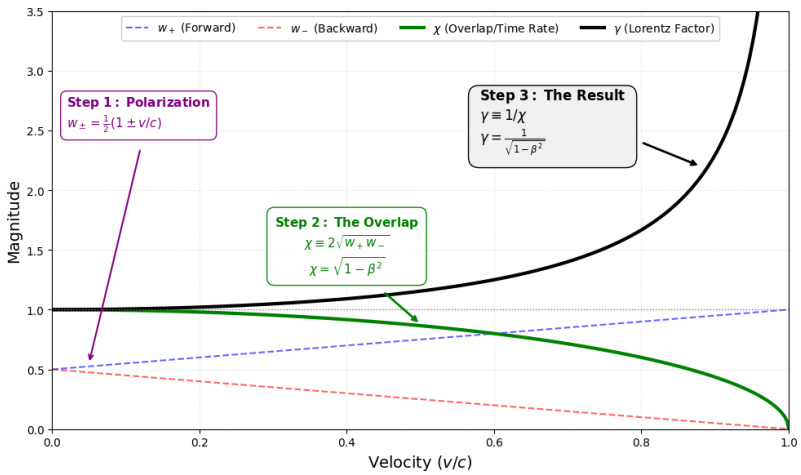


B. Propagation: Light propagates along null directions; measurement selects a realized direction.



C. Gravity: Mass reduces directional accessibility, inducing null focusing (an "entropic pressure" interpreted macroscopically as gravitational attraction).

2. Deriving Relativity from Statistics



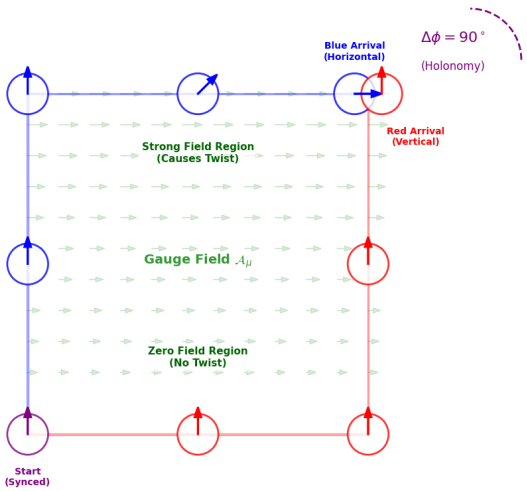
Velocity is defined as a bias in the directional sampling weights $w_\pm$.

$$w_\pm = \frac{1}{2}(1 \pm v/c)$$

The symmetric overlap (χ) is constitutively identified with the operational clock rate in the reduced directional sector. As $v \to c$, $\chi \to 0$:

$$\gamma \equiv \frac{1}{\chi} = \frac{1}{\sqrt{1 - v^2/c^2}}$$

3. Geometric Origin of Force (Holonomy)



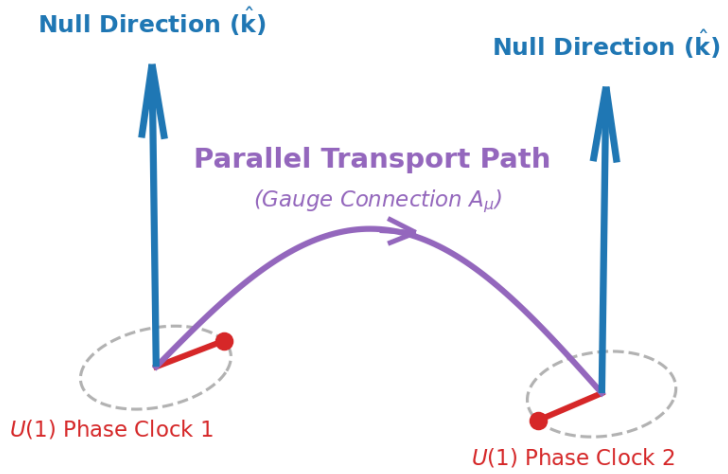
Gauge interactions are geometric phase twists (Holonomies) arising from the spinor representation of directions.

Transporting a state through the substrate requires a connection $\mathcal{A}_\mu$ to maintain a consistent internal phase ("clock") across different frames.

$$\text{Phase shift} \propto \oint \mathcal{A}_\mu dx^\mu$$

This identifies forces not as independent postulates, but as kinematic necessities of directional transport.

4. Emergent Connections: Gravity & Electromagnetism



Classical gauge and gravitational fields need not be introduced as independent substances; they arise as mandatory geometric connections required to compare local substrate states.

Because the absolute internal state of the null-directional ensemble is unobservable, consistent comparison across spacetime requires a local synchronization rule (a connection field):

- **Electromagnetism (A_μ):** The $U(1)$ connection tracking spinor phase redundancy.
- **Gravity (ω_μ^{ab}):** The $SO(1,3)$ connection tracking local Lorentz frame redundancy.

Under standard locality, covariance, and low-derivative effective closure assumptions, the familiar Maxwell equations and Einstein-type macroscopic gravitational closure emerge as the leading effective dynamics of these connections.

5. The "Classical" Wall: A Testable Discriminator

In ideal isolation, standard quantum mechanics predicts persistent coherence ($V = V_0$).

Finite-Throughput Benchmark Prediction: If the underlying directional substrate has finite effective informational throughput, the overlap-based clock functional may exhibit an intrinsic stochastic component, producing phase diffusion even in the absence of environmental coupling.

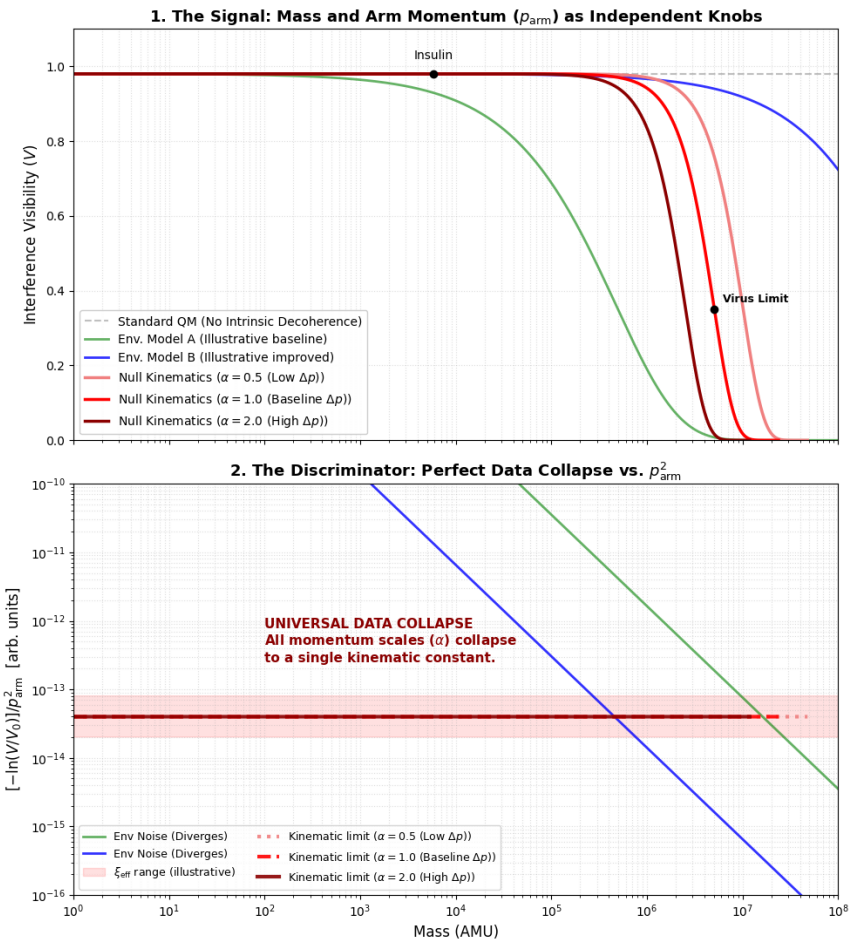
Benchmark Discriminator: In the minimal stochastic benchmark, the induced decoherence rate scales with the square of the interferometer arm momentum,

$$\Gamma_\phi \propto p_{\text{arm}}^2.$$

The arm momentum is parameterized as $p_{\text{arm}} = \alpha p_0$. In transport-defined regimes ($p_{\text{arm}} \sim m v_{\text{arm}}$), the scaling factor α represents relative arm velocity (v_{arm}/v_0), yielding quadratic scaling in both mass and velocity. In recoil-defined optical regimes, α strictly corresponds to the Large Momentum Transfer (LMT) photon order.

Illustrative Benchmarks:

- **Insulin (5.8k amu):** Observed interference.
- **Viruses ($\sim 10^6$ amu):** Proposed transition-scale target (illustrative).



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Note: This visual guide is for illustrative purposes. All equations, theorems, and proofs are formally derived in the following work:

Thompson, E. (2026). Null-Directional Kinematics and the Statistical Structure of Spacetime: Clock Functionals, Transport Geometry, and Directional Support Structure. Zenodo.
https://doi.org/10.5281/zenodo.19226343